

Diffusion-Based Local Image Deblurring

Jinzi Luo (UNI: jl7199)

1 Overview of the Field

Image deblurring aims to recover sharp images from blurry observations. Most methods target *global* blur caused by camera shake, progressing from multi-scale CNNs [1] to Transformers [2] and state-space models [3], achieving strong results on benchmarks like GoPro and RealBlur. However, *local* motion blur—where only moving objects are blurred while the background stays sharp—is far more common in real-world photography (e.g., capturing pedestrians, vehicles, sports scenes) yet remains underexplored. Directly applying global deblurring models to locally blurred images is problematic: they lack awareness of blur region boundaries, tend to unnecessarily modify sharp background areas, and are trained on a different data distribution where blur is uniform.

Li et al. [4] first formalized this task by introducing ReLoBlur, a real-world local motion blur dataset of $\sim 2K$ paired images captured via a beam-splitting system, along with a Local Blur-Aware Gated network (LBAG) that uses blur detection and gating mechanisms. Follow-up works include adaptive window pruning Transformers [8] that selectively process blurred regions and parameterized blur kernel priors [9] that estimate per-region motion parameters for guided restoration. On the generative side, diffusion-based restoration has advanced rapidly: DiffBIR [5] leverages Stable Diffusion priors with a modified ControlNet for blind image restoration, while DPIR [10] explores DiT-based architectures with dual prompting, and VIDID [6] introduces visual-instructed degradation diffusion for all-in-one restoration. Despite these advances, no prior work combines diffusion models with multi-condition spatial control specifically for local deblurring, nor addresses the critical data scarcity through systematic synthetic augmentation.

2 Problem Formulation

Given an image I_b with locally blurred regions, a blur mask M , and optionally a segmentation map S , produce a restored image $\hat{I}_s$ that: (1) restores blurred regions to match ground truth I_s , (2) preserves the sharp background, and (3) ensures natural boundary transitions.

3 Proposed Approach

I will build a diffusion-based local deblurring system using Stable Diffusion + ControlNet [7], extending prior work [4]. The approach has three components:

(a) Multi-condition ControlNet architecture. The blurred region is passed to ControlNet as a spatial condition, while a CLIP image encoder provides semantic guidance in place of text prompts. Additional conditions—the input image (background context), blur mask, and object segmentation map—are concatenated with the diffusion latent for region-aware restoration.

(b) Synthetic data pipeline for pre-training. To address ReLoBlur’s data scarcity ($\sim 2K$ images), I will: (i) apply blur kernels to COCO images using instance segmentation masks, (ii) augment global blur datasets (GoPro, RealBlur) with arbitrary-shape and object-aware masks. A mask-centered cropping strategy unifies all datasets to 512×512 while preserving aspect ratios.

(c) Progressive training strategy. Pre-train on synthetic + global blur data, then fine-tune on ReLoBlur with local masks. I will explore improved sampling (initializing from the blurred input latent rather than pure noise) and repaint-based post-processing.

4 Deliverables, Evaluation, and Compute Plan

Quantitative: PSNR and SSIM in three modes (basic, weighted, aligned) on ReLoBlur test set; target: match or exceed LBAG baseline (PSNR ≈ 34.71 , SSIM ≈ 0.967). **Qualitative:** Visual comparisons and ablation studies on mask conditioning, fine-tuning, synthetic data volume, and denoising steps. **Code:** Training and inference scripts will be provided. **Compute:** Cloud GPU (single NVIDIA A100 40GB) with pre-trained SD/ControlNet checkpoints; ~ 50 – 80 GPU-hours total. Google Colab Pro+ as backup.

References

- [1] S. Nah, T.H. Kim, K.M. Lee, “Deep multi-scale convolutional neural network for dynamic scene deblurring,” CVPR, 2017.
- [2] S.W. Zamir et al., “Restormer: Efficient transformer for high-resolution image restoration,” CVPR, 2022.
- [3] L. Kong et al., “Efficient visual state space model for image deblurring,” CVPR, 2025.
- [4] H. Li et al., “Real-world deep local motion deblurring,” AAAI, 2023.
- [5] X. Lin et al., “DiffBIR: Towards blind image restoration with generative diffusion prior,” ECCV, 2024.
- [6] Z. Luo et al., “Visual-instructed degradation diffusion for all-in-one image restoration,” CVPR, 2025.
- [7] L. Zhang, A. Rao, M. Agrawala, “Adding conditional control to text-to-image diffusion models,” ICCV, 2023.
- [8] I. Kim et al., “Local motion deblurring via adaptive window pruning transformer,” CVPR, 2024.
- [9] X. Fang et al., “Parameterized blur kernel prior learning for local motion deblurring,” CVPR, 2025.
- [10] Z. Chen et al., “Dual prompting image restoration with diffusion transformers,” CVPR, 2025.